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Tab detection mechanism and tab detection device

Abstract

This application discloses a tab detection mechanism and a tab detection device. The tab detection mechanism may be configured to detect a tab portion on an electrode core. The tab detection mechanism may include a visual camera; a prism disposed below the visual camera and configured to reflect light on a side surface of the tab portion to the visual camera; and a first driver, connected to the visual camera and configured to drive the visual camera to move in a vertical direction. The visual camera is capable of shooting a plurality of images while moving.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2023/084767, filed Mar. 29, 2023, which claims priority to Chinese Patent Application No. 202223105124.5, filed with the China National Intellectual Property Administration on Nov. 22, 2022 and entitled “TAB DETECTION MECHANISM AND TAB DETECTION DEVICE”, each of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

(1) This application relates to the field of battery production, and in particular, to a tab detection mechanism and a tab detection device.

BACKGROUND

(2) Energy conservation and emission reduction are crucial to the sustainable development of automobiles. Electric vehicles have drawn wide attention due to their advantage of environment friendliness. Traction battery is an important part of electric vehicles.

(3) An electrode core of a traction battery is provided with a tab portion. During production of the traction battery, it is usually necessary to detect the tab portion so as to identify the state of the tab portion. In some cases, the tab portion is mainly detected by means of artificial vision. This results in low detection efficiency and easily leads to false detection, affecting the quality of batteries.

SUMMARY

(4) In view of this, embodiments of this application provide a tab detection mechanism and a tab detection device, capable of detecting a state of a tab portion automatically, thereby solving the problems about low detection efficiency and probable false detection.

(5) An embodiment of a first aspect of this application provides a tab detection mechanism, configured to detect a tab portion on an electrode core, including: a visual camera; a prism, disposed below the visual camera and configured to reflect light on a side surface of the tab portion to the visual camera; and a first drive member or driver, connected to the visual camera and configured to drive the visual camera to move in a vertical direction, wherein the visual camera is capable of shooting a plurality of images while moving.

(6) The tab detection mechanism provided by the embodiment of this application includes the visual camera, the prism and the first drive member, the prism can reflect the light on the side surface of the tab portion to the visual camera, and the first drive member can drive the visual camera to move in the vertical direction. Therefore, the above tab detection mechanism can automatically shoot an image of the side surface of the tab portion, so that the state of the tab

portion is detected and whether the tab portion is abnormal or not is analyzed based on the image, solving the problems about low detection efficiency and probable false detection caused in manual detection, thereby improving the detection efficiency and accuracy. In addition, if a space on two sides of the tab portion is insufficient, the visual camera cannot be placed horizontally. According to the embodiment of this application, a view field direction of the visual camera is changed by virtue of the prism, and the visual camera can be placed vertically or obliquely on an outer side of the tab portion, thereby solving the problem that the visual camera is inconvenient to arrange. Moreover, the first drive member can drive the visual camera to implement fly-shot imaging, so that the visual camera can acquire a complete image of the side surface of the tab portion during one-time image acquisition, further improving the detection efficiency and guaranteeing a detection effect.

(7) In some embodiments, the visual camera includes a first bracket connected to the first drive member, as well as a camera and a telecentric lens that are disposed on the first bracket, and the telecentric lens and the camera are sequentially disposed above the prism in the vertical direction.

(8) With use of the foregoing technical solution, the prism can change the view field direction of the visual camera. Therefore, the camera and the telecentric lens can both be placed vertically, solving the problem that the camera is inconvenient to place due to a too small space on sides of the tab portion. The first drive member can drive the camera and the telecentric lens to move in the vertical direction simultaneously, thereby realizing fly-shot imaging of the camera.

(9) In some embodiments, an included angle between the prism and the vertical direction is 45° .

(10) With use of the 45° prism, the view field direction of the camera can be changed by virtue of the prism, the prism can reflect light on the side surface of the tab portion, and the camera acquires an image of the side surface of the tab portion by shooting a surface of the prism.

(11) In some embodiments, a mounting height of the telecentric lens from the prism is greater than or equal to a difference between a working distance of the telecentric lens and a horizontal distance from an upper edge of the prism to the tab portion.

(12) With the above condition satisfied, the prism can be installed on one side of the tab portion, and the mounting height of the telecentric lens depends on the working distance of the telecentric lens and the horizontal distance from the upper edge of the prism to the tab portion. In this way, it can be ensured that the side surface of the tab portion is within the working distance of the telecentric lens, and thus the visual camera can shoot a complete and clear image of the side surface of the tab portion. The above tab detection mechanism can detect a single layer of tabs or a plurality of layers of tabs and can detect situations such as unevenness of the plurality of layers of tabs and staggering of the tabs.

(13) In some embodiments, the tab portion includes a plurality of tabs stacked in a first direction, a moving distance of the visual camera in the vertical direction is greater than a maximum offset distance of the plurality of tabs in a second direction, and the second direction is perpendicular to the first direction.

(14) With the above condition satisfied, the visual camera moves a certain distance in the vertical direction, so that it can be ensured that the visual camera can shoot side surfaces of the plurality of tabs and shoot staggered positions of the tabs, thereby improving the detection accuracy.

(15) In some embodiments, a maximum height difference between the upper edge of the prism and a lower edge of the prism is greater than a view field of the visual camera, and the view field of the visual camera is greater than a total thickness of the tab portion.

(16) With the above condition satisfied, the prism can be ensured to reflect the entire side surface of the tab portion and the visual camera can be ensured to shoot the entire side surface of the tab portion.

(17) In some embodiments, the tab detection mechanism further includes a second drive member or driver, and the second drive member is connected to the prism and configured to drive the prism to move in a direction close to or away from the electrode core.

(18) With the second drive member provided, the prism is enabled to move in the direction close to or away from the electrode core so as to keep away from the electrode core that is being conveyed.

(19) In some embodiments, the second drive member is a cylinder, and the second drive member is configured to drive the prism to move in the vertical direction.

(20) With use of the foregoing technical solution, the prism moves in only the vertical direction and can be placed at a side portion of the tab portion after moving to a proper position, thereby ensuring that the prism can reflect light on the side surface of the tab portion.

(21) An embodiment of a second aspect of this application provides a tab detection device, configured to detect a first tab portion and a second tab portion on an electrode core, including: a first tab detection mechanism or detector, configured to detect an inner side surface of the first tab portion; a second tab detection mechanism or detector, configured to detect an outer side surface of the second tab portion; a third tab detection mechanism or detector, configured to detect an outer side surface of the first tab portion; and a fourth tab detection mechanism or detector, configured to detect an inner side surface of the second tab portion, wherein

(22) at least one of the first tab detection mechanism, the second tab detection mechanism, the third tab detection mechanism and the fourth tab detection mechanism is the tab detection mechanism according to the first aspect.

(23) The tab detection device includes the first tab detection mechanism, the second tab detection mechanism, the third tab detection mechanism and the fourth tab detection mechanism that are capable of detecting the inner side surface of the first tab portion, the outer side surface of the first tab portion, the inner side surface of the second tab portion and the outer side surface of the second tab portion, respectively. The above tab detection device can implement automatic and comprehensive detection of tab portions without manual operation, thereby improving the detection efficiency and accuracy and improving the quality of batteries.

(24) In some embodiments, a prism in the second tab detection mechanism and a prism in the third tab detection mechanism are located on a same second bracket and are arranged opposite to each other. A prism in the first tab detection mechanism and a prism in the fourth tab detection mechanism are arranged in a mirror-image form.

(25) In this way, the second tab detection mechanism and the third tab detection mechanism can implement synchronous detection, further improving the detection efficiency.

(26) In some embodiments, the tab detection device further includes a light shield plate and a third drive member or driver, the third drive member is configured to drive the light shield plate to move, so that the light shield plate is located between the prism of the second tab detection mechanism and the prism of the third tab detection mechanism, and the light shield plate is a non-transparent plate.

(27) With use of the foregoing technical solution, the light shield plate can prevent mutual interference between the second detection mechanism and the third detection mechanism, and the light shield plate is movable so as not to hinder conveying of the electrode core.

(28) In some embodiments, the light shield plate is a non-transparent plate subjected to matte black processing.

(29) With use of the foregoing technical solution, the light shield plate is subjected to matte black processing, therefore the light shield plate can not only shield light but also avoid light reflection, thereby preventing reflected light from affecting an imaging effect of the visual camera.

(30) In some embodiments, the tab detection device further includes a conveying mechanism or conveyor, and the conveying mechanism is configured to convey the electrode core, so that the electrode core sequentially passes through the first tab detection mechanism, the second tab detection mechanism, the third tab detection mechanism and the fourth tab detection mechanism.

(31) With use of the foregoing technical solution, the conveying mechanism can drive an electrode core positioning table and the electrode core to move, so that the electrode core sequentially passes through the four tab detection mechanisms for detecting the inner and outer side surfaces of the

first tab portion and the second tab portion. In this way, wrinkling, folding, cracking and the like of the tabs, as well as staggering situation and evenness of a plurality of layers of tabs are detected. (32) The foregoing description is merely an overview of the technical solution of this application. For a better understanding of the technical means in this application, they can be implemented according to the content of the specification, and to make the above and other objectives, features and advantages of this application more obvious and easier to understand, the following describes specific implementations of this application.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) To describe the technical solutions in the embodiments of this application more clearly-, the following briefly describes the accompanying drawings required for describing the embodiments or conventional technologies. Apparently, the accompanying drawings in the following description show merely some of the embodiments of this application, and persons of ordinary skill in the art may still derive other drawings from these accompanying drawings without creative efforts.

(2) FIG. 1 is a front view of an electrode core according to an embodiment of this application,

(3) FIG. 2 is a top view of an electrode core according to an embodiment of this application;

(4) FIG. 3 is a three-dimensional schematic diagram of a tab detection mechanism according to an embodiment of this application;

(5) FIG. 4 is a three-dimensional schematic diagram of a visual camera and a first drive member in the tab detection mechanism shown in FIG. 3;

(6) FIG. 5 is a three-dimensional schematic diagram of a prism assembly in the tab detection mechanism shown in FIG. 3;

(7) FIG. 6 is a schematic diagram showing a principle of the tab detection mechanism shown in FIG. 3;

(8) FIG. 7 is a locally enlarged view of part A in the tab detection mechanism shown in FIG. 6;

(9) FIG. 8 is a locally enlarged view of part B in the tab detection mechanism shown in FIG. 6;

(10) FIG. 9 is a three-dimensional schematic diagram of a tab detection device according to an embodiment of this application;

(11) FIG. 10 is a three-dimensional schematic diagram of a second tab detection mechanism, a third tab detection mechanism, a light shield plate and an electrode core in the tab detection device shown in FIG. 9;

(12) FIG. 11 is a three-dimensional schematic diagram of two prisms, a second bracket and a second drive member in the tab detection device shown in FIG. 9; and

(13) FIG. 12 is a three-dimensional schematic diagram of a light shield assembly in the tab detection device shown in FIG. 9.

(14) The reference numerals in the drawings refer to: **100**. tab detection mechanism; **10**. visual camera; **11**. first bracket; **111**. fixing plate; **112**. camera stand; **113**. lens holder; **12**. camera; **13**. telecentric lens; **20**. prism assembly; **21**. prism; **22**. second bracket; **23**. second drive member; **30**. first drive member, **40**. light shield plate, **50**. third drive member, **100a**. first tab detection mechanism; **100b**. second tab detection mechanism; **100c**. third tab detection mechanism; **100d**. fourth tab detection mechanism; **200**. electrode core; **210**. tab portion; **210a**. first tab portion; **210b**. second tab portion; **300**. conveying mechanism; **310**. electrode core positioning table; and **1000**. tab detection device.

DETAILED DESCRIPTION OF EMBODIMENTS

(15) The following describes in detail the embodiments of technical solutions of this application with reference to the accompanying drawings. The following embodiments are merely intended for a clearer description of the technical solutions of this application and therefore are used as just

examples which do not constitute any limitation on the protection scope of this application.

(16) Unless otherwise defined, all technical and scientific terms used herein shall have the same meanings as commonly understood by those skilled in the art to which this application relates. The terms used herein are intended to merely describe the specific embodiments rather than to limit this application. The terms “include”, “comprise”, and any other variations thereof in the specification, claims and brief description of drawings of this application are intended to cover non-exclusive inclusions.

(17) In the description of the embodiments of this application, the terms “first”, “second” and the like are merely intended to distinguish different objects, and shall not be understood as any indication or implication of relative importance or any implicit indication of the number, particular sequence or primary-secondary relationship of the technical features indicated. In the description of the embodiments of this application, “a plurality of” means more than two unless otherwise specifically stated.

(18) Herein, reference to “embodiment” means that specific features, structures or characteristics described with reference to the embodiment may be incorporated in at least one embodiment of this application. The word “embodiment” appearing in various places in the specification does not necessarily refer to the same embodiment or an independent or alternative embodiment that is exclusive of other embodiments. It is explicitly or implicitly understood by those skilled in the art that the embodiments described herein may be combined with other embodiments.

(19) In the description of the embodiments of this application, the term “and/or” is merely for describing an associative relationship of associated objects, indicating that three relationships may be present. For example, A and/or B may indicate the following three cases: presence of only A, presence of both A and B, and presence of only B. In addition, the character “I” herein generally indicates an “or” relationship between contextually associated objects.

(20) In the description of the embodiments of this application, the term “a plurality of” means more than two (inclusive). Similarly, “a plurality of groups” means more than two (inclusive) groups, and “a plurality of pieces” means more than two (inclusive) pieces.

(21) In the description of the embodiments of this application, the orientations or positional relationships indicated by the technical terms “center”, “longitudinal”, “transverse”, “length”, “width”, “thickness”, “upper”, “lower”, “front”, “rear”, “vertical”, “horizontal”, “top”, “bottom”, “inside”, “outside”, and the like are based on the orientations or positional relationships as shown in the accompanying drawings. These terms are merely for ease and brevity of description of the embodiments of this application rather than indicating or implying that the apparatuses or components mentioned must have specific orientations or must be constructed or manipulated according to specific orientations, and therefore shall not be construed as limitations on the embodiments of this application.

(22) In the description of the embodiments of this application, unless otherwise specified and defined explicitly, the terms “mount”, “connect”, “join”, and “fix” should be understood in their general senses. For example, they may refer to a fixed connection, a detachable connection, or an integral connection, may refer to a mechanical connection or an electrical connection, and may refer to a direct connection, an indirect connection via an intermediate medium, internal communication between two elements, or an interaction relationship of two elements. Persons of ordinary skill in the art can understand specific meanings of these terms in this application as appropriate to specific situations.

(23) Currently, energy conservation and emission reduction are crucial to the sustainable development of automobiles. Electric vehicles have drawn wide attention due to their advantage of environment friendliness. Traction battery is an important part of electric vehicles.

(24) Traction battery may include, but is not limited to, three forms: battery cell, battery module and battery pack. Battery cell is the smallest unit of a traction battery and typically includes a housing, a cover plate, an electrode core, an electrolyte capsule, and other functional components.

In application, a plurality of battery cells can be connected by using an electric connecting structure and a fixing structure, and then are assembled with a thermal management system to form a battery, and the battery can be directly assembled on an electric apparatus to provide electrical energy for the electric apparatus. Alternatively, a plurality of battery cells are first assembled into battery modules by using an electric connecting structure, a fixing structure and the like, and then a plurality of battery modules, fitting with a thermal management system, are assembled on an electric apparatus to provide electrical energy for the electrical energy. Still alternatively, a plurality of battery modules, fitting with a thermal management system, are assembled into a box to form a battery pack, and then the battery pack is directly assembled on an electric apparatus to provide electrical energy for the electric apparatus.

(25) An electrode core of a battery cell is provided with tab portions. The tab portions are metal conductors for leading positive and negative electrodes out of the electrode core. Two tab portions extend out of one electrode core. The tab portions serve as contact points during charge and discharge of the battery. In addition, each tab portion typically includes a plurality of layers of tabs, to be specific, the tab portion includes a plurality of stacked tabs.

(26) During production of a traction battery, it is usually necessary to detect the tab portion so as to identify the state of the tab portion. In some cases, the tab portion is mainly detected by means of artificial vision. This results in low detection efficiency and easily leads to false detection, affecting the quality of batteries.

(27) To solve the problems, the applicants have designed a tab detection mechanism and a tab detection device through in-depth research. The tab detection mechanism includes a visual camera, a prism and a first drive member. The prism can reflect light on a side surface of a tab portion to the visual camera, so that the visual camera can acquire an image of the side surface of the tab portion. In addition, the first drive member can drive the visual camera to move in a vertical direction, and the visual camera can shoot a plurality of images while moving. Therefore, the above tab detection mechanism can acquire a complete image of the side surface of the tab portion with a one-time image acquisition process, thereby implementing automatic visual detection for the tab portion, solving the problems about low detection efficiency and probable false detection for the tab portion, and guaranteeing the quality of batteries.

(28) An embodiment of a first aspect of this application provides a tab detection mechanism configured to detect a tab portion on an electrode core.

(29) FIG. 1 is a front view of an electrode core **200** in an embodiment, and FIG. 2 is a top view of an electrode core **200** in an embodiment. Referring to FIG. 1 and FIG. 2, the electrode core **200** is provided with two tab portions **210**, where each tab portion **210** includes a plurality of tabs stacked in a first direction, and the first direction is a thickness direction T of the electrode core **200**. In the electrode core **200** shown in FIG. 1, the plurality of tabs in each tab portion **210** are uneven. In the electrode core **200** shown in FIG. 2, the plurality of tabs in each tab portion **210** are staggered, that is, the plurality of tabs are staggered in a second direction perpendicular to the first direction, and the second direction is a length direction L of the electrode core **200**.

(30) The two tab portions **210** are a first tab portion **210a** and a second tab portion **210b**, respectively, where the first tab portion **210a** is an anode tab, and the second tab portion **210b** is a cathode tab.

(31) The tab detection mechanism provided in this application can be used to detect the situation of the plurality of layers of tabs, but is not limited thereto. The tab detection mechanism may also be used to detect the situation of a single layer of tabs.

(32) Referring to FIG. 3, the tab detection mechanism **100** includes a visual camera **10**, a prism **21** and a first drive member **30**. The prism **21** is disposed below the visual camera **10** and configured to reflect light on a side surface of the tab portion **210** to the visual camera **10**; the first drive member **30** is connected to the visual camera **10** and configured to drive the visual camera **10** to move in a vertical direction; and the visual camera **10** is capable of shooting a plurality of images

while moving.

(33) The visual camera **10** can acquire an image projected to a sensor through a lens. The visual cameras **10** can be classified into CCD cameras and CMOS cameras according to the types of chips, and can be classified into line-array cameras and area-array cameras according to structural characteristics of the sensors. The type of the visual camera **10** is not limited in this application.

(34) The prism **21** includes a reflective surface, and the reflective surface of the prism **21** is toward the side surface of the tab portion **210** so as to reflect light on the side surface of the tab portion **210**. The reflective surface of the prism **21** may also be disposed toward the visual camera **10**, and the visual camera **10** can shoot the reflective surface of the prism **21**, so as to acquire an image of the side surface of the tab portion **210**.

(35) In particular, the “side surface” of the tab portion **210** may be an inner side surface or an outer side surface of the tab portion **210**, where the “inner side surface” refers to a side surface of a side of one tab portion **210** close to another tab portion **210**, and the “outer side surface” refers to a side surface of a side of one tab portion **210** facing away from another tab portion **210**.

(36) Optionally, the visual camera **10** is disposed in the vertical direction, and the prism **21** is inclined with respect to the vertical direction, so that the prism **21** can be toward both the side surface of the tab portion **210** and the visual camera **10**, thereby helping to reflect the light on the side surface of the tab portion **210** to the visual camera **10**. It can be understood that the visual camera **10** may also be disposed in a manner of forming a specified included angle with respect to the vertical direction as long as the prism **21** can reflect the light on the side surface of the tab portion **210** to the visual camera **10**.

(37) The first drive member **30** is configured to drive the visual camera **10** to move in the vertical direction. The visual camera **10** implements fly-shot imaging while moving, such that the visual camera **10** can continuously acquire a plurality of images of the tab portion **210**, thereby helping to acquire a complete image of the tab portion **210**. The first drive member **30** may be, but is not limited to, a motor ball screw module. The first drive member **30** may also be a cylinder or another drive member.

(38) Detection principle of the tab detection mechanism **100** is as follows: The electrode core **200** is fixed to an electrode core positioning table **310**; the prism **21** is disposed on one side of the electrode core positioning table **310**; the visual camera **10** is disposed above the prism **21**; the prism **21** projects the image of the side surface of the tab portion **210** onto the visual camera **10**; the first drive member **30** drives the visual camera **10** to move in the vertical direction; and the visual camera **10** continuously shoots a plurality of images while moving so as to acquire a complete image of the side surface of the tab portion **210**, so that the state of the tab portion **210** can be detected and whether the tab portion **210** is abnormal or not is determined based on the image.

(39) The tab portion **210** includes one or a plurality of tabs. In a case that the tab portion **210** includes one tab, the visual camera **10** can acquire an image of the side surface of the tab for detecting whether the tab is in abnormal situations such as wrinkling, folding and cracking. In a case that the tab portion **210** includes a plurality of stacked tabs, the visual camera **10** can acquire images of the plurality of tabs for detecting whether the number of the tabs is correct or not, whether the tabs are folded or not, whether the evenness of the plurality of layers of tabs is qualified or not, whether staggered distances of the plurality of layers of tabs are qualified or not, and the like.

(40) The tab detection mechanism **100** provided by this embodiment of this application includes the visual camera **10**, the prism **21** and the first drive member **30**, where the prism **21** can reflect the light on the side surface of the tab portion **210** to the visual camera **10**, and the first drive member **30** can drive the visual camera **10** to move in the vertical direction. Therefore, the tab detection mechanism **100** can automatically shoot an image of the side surface of the tab portion **210**, so that the state of the tab portion **210** is detected and whether the tab portion **210** is abnormal or not is analyzed based on the image, solving the problems about low detection efficiency and probable

false detection caused in manual detection, thereby improving the detection efficiency and accuracy.

(41) In addition, if a space on two sides of the tab portion **210** is insufficient, the visual camera **10** cannot be placed horizontally. According to the embodiment of this application, a view field direction of the visual camera **10** is changed by virtue of the prism **21**, and the visual camera **10** can be placed vertically or obliquely on an outer side of the tab portion **210**, thereby solving the problem that the visual camera **10** is inconvenient to arrange. Moreover, the first drive member **30** can drive the visual camera **10** to implement fly-shot imaging, so that the visual camera **10** can acquire a complete image of the side surface of the tab portion **210** during one-time image acquisition, further improving the detection efficiency and guaranteeing the detection effect.

(42) Referring to FIG. 3 and FIG. 4, in some embodiments, the visual camera **10** includes a first bracket **11** connected to the first drive member **30**, as well as a camera **12** and a telecentric lens **13** that are disposed on the first bracket **11**, where the telecentric lens **13** and the camera **12** are sequentially disposed above the prism **21** in the vertical direction.

(43) The first bracket **11** is connected to the first drive member **30**, such that the first drive member **30** can drive the camera **12** and the telecentric lens **13** to move in the vertical direction by virtue of the first bracket **11**. The first bracket **11** may include a fixing plate **111** disposed vertically, and a camera stand **112** and a lens holder **113** that are disposed on one side of the fixing plate **111**, where the camera **12** is disposed on the fixing plate **111** through the camera stand **112**, and the telecentric lens **13** is disposed on the fixing plate **111** through the lens holder **113**. The first drive member **30** is connected to one side of the fixing plate **111** facing away from the camera stand **112** and the lens holder **113**.

(44) The camera **12** images the side surface of the tab portion **210** by reflection of the telecentric lens **13** and the prism **21**. The telecentric lens **13** (Telecentric) is designed mainly for correcting parallax of conventional industrial lenses. The telecentric lens enables an acquired image to be unchanged in magnifying power within a specified object distance range. This is a quite important application for a case in which measured objects are not on a same object surface. The telecentric lens **13** has unique optical characteristics such as high resolution, ultra-wide depth of field, ultra-low distortion and unique parallel light design, and therefore can meet the requirements of machine vision precision detection.

(45) With use of the foregoing technical solution, the prism **21** can change a view field direction of the visual camera **10**. Therefore, the camera **12** and the telecentric lens **13** can both be placed vertically, solving the problem that the camera **12** is inconvenient to place due to a too small space on sides of the tab portion **210**. The first drive member **30** can drive the camera **12** and the telecentric lens **13** to move in the vertical direction simultaneously, thereby realizing fly-shot imaging of the camera **12**.

(46) Referring to FIG. 1 and FIG. 3, in some embodiments, an included angle between the prism **21** and the vertical direction is 45° .

(47) With use of the prism **21** in 45° from the vertical direction, the view field direction of the camera **12** can be changed, the prism **21** can reflect light on the side surface of the tab portion **210**, and the camera **12** acquires an image of the side surface of the tab portion **210** by shooting a surface of the prism **21**. In addition, the 45° prism **21** can reflect light and make incident light be deviated 90° , so that the vertically placed visual camera **10** shoots the tab portion **210** that is placed horizontally.

(48) It can be understood that in other embodiments, the angle of the prism **21** is not limited to 45° , provided that the prism **21** can project the image of the side surface of the tab portion **210** to the visual camera **10**. For example, the angle of the prism **21** is 30° to 60° .

(49) Referring to FIG. 3 to FIG. 8, in some embodiments, a mounting height of the telecentric lens **13** from the prism **21** is greater than or equal to a difference between a working distance D of the telecentric lens **13** and a horizontal distance L2 from an upper edge of the prism **21** to the tab

portion **210**.

(50) A to-be-tested electrode core **200** is placed flat on the electrode core positioning table **310**, that is, a surface, defined by width and length, of the electrode core **200** is placed flat on the electrode core positioning table **310**, and two tab portions **210** are disposed on one side of the electrode core **200**.

(51) The working distance D of the telecentric lens **13** satisfies a condition: $D=L1+L2$.

(52) The working distance D of the telecentric lens **13** is related to type selection of the telecentric lens **13**, and is an intrinsic parameter of the telecentric lens **13**; $L1$ represents a distance from an uppermost light reflection point on the prism **21** to the telecentric lens **13**; and $L2$ represents a horizontal distance from the upper edge of the prism **21** to the tab portion **210**, that is, a distance from the upper edge of the prism **21** to an outermost edge of the tab portion **210** in a horizontal direction, where the upper edge of the prism **21** refers to an upper edge of a surface of the prism **21** in the vertical direction.

(53) In particular, the uppermost reflection point on the prism **21** refers to a highest point, capable of reflecting light, on the prism **21** in the vertical direction. There may be tabs staggered on the side surface of the tab portion **210**, so the outermost edge of the tab portion **210** refers to an outermost edge of a longest tab.

(54) Based on the above condition of the working distance D , it can be obtained that $L1=D-L2$.

(55) Further, the mounting height of the telecentric lens **13** from the prism **21** is greater than or equal to $L1$, that is, the mounting height of the telecentric lens **13** from the prism **21** is greater than or equal to the difference between the working distance D of the telecentric lens **13** and the horizontal distance $L2$ from the upper edge of the prism **21** to the tab portion **210**. The mounting height of the telecentric lens **13** from the prism **21** refers to a height of a lower edge of the telecentric lens **13** with respect to the prism **21**. The lower edge of the telecentric lens **13** refers to the extreme edge of one side of the telecentric lens **13** toward the prism **21**.

(56) Optionally, the mounting height of the telecentric lens **13** from the prism **21** is greater than $L1$. that is, the mounting height of the telecentric lens **13** from the prism **21** is greater than the difference between the working distance D of the telecentric lens **13** and the horizontal distance $L2$ from the upper edge of the prism **21** to the tab portion **210**, thereby guaranteeing a certain shooting allowance.

(57) After the electrode core **200** is wound, the plurality of layers of tabs are in different states. The tabs in the respective layers are different in evenness and wrinkled, gaps are formed between the tab layers, and different thicknesses of different tabs of the electrode core **200** result in different shooting ranges of the camera. Please also refer to FIG. 1 to show the situation about unevenness of the plurality of layers of tabs; please refer to FIG. 2 to show the situation about staggering of the plurality of layers of tabs.

(58) With the above condition satisfied, the prism **21** can be installed on one side of the tab portion **210** and toward the side surface of the tab portion **210**, and the mounting height of the telecentric lens **13** depends on the working distance of the telecentric lens **13** and the horizontal distance from the upper edge of the prism **21** to the tab portion **210**. In this way, it can be ensured that the side surface of the tab portion **210** is within the working distance of the telecentric lens **13**, and thus the visual camera **10** can shoot a complete and clear image of the side surface of the tab portion **210**. The tab detection mechanism **100** can detect a single layer of tabs or a plurality of layers of tabs and can detect situations such as unevenness of the plurality of layers of tabs and staggering of the tabs.

(59) In some embodiments, the tab portion **210** includes a plurality of tabs stacked in a first direction. A moving distance X of the visual camera **10** in the vertical direction is greater than a maximum offset distance H of the plurality of tabs in a second direction, where the second direction is perpendicular to the first direction, satisfying the following condition: $X>H$.

(60) Specifically, the first direction is a thickness direction T of the electrode core **200**, and the

second direction is a length direction L of the electrode core **200**. The plurality of tabs are stacked in the first direction and are likely to be staggered in the second direction. Therefore, during detection, it is necessary to detect an offset distance of the plurality of tabs.

(61) The moving distance X of the visual camera **10** in the vertical direction is a moving distance of the visual camera **10** driven by the first drive member **30**. The maximum offset distance of the plurality of tabs in the second direction refers to an offset length of a longest tab and a shortest tab when viewed from one side of the tab portion **210**. Optionally, the moving distance X is 20 mm to 30 mm, but is not limited thereto, which is specifically determined based on the tab portion **210**.

(62) With the above condition satisfied, the visual camera **10** moves a certain distance in the vertical direction, so that it can be ensured that the visual camera **10** can shoot side surfaces of the plurality of tabs and shoot staggered positions of the tabs, thereby improving the detection accuracy.

(63) It can be understood that in order to ensure a certain allowance, the moving distance of the visual camera **10** is greater than or equal to the sum of the maximum offset distance H of the plurality of tabs and the allowance. The allowance may include allowance a and allowance b, where allowance a is an allowance of a distance from the upper edge of the prism **21** to the longest tab, and allowance b is an allowance of a distance from the upper edge of the prism **21** to the shortest tab.

(64) Therefore, the moving distance of the visual camera **10** satisfies the following condition: $X \geq a + H + b$, thereby ensuring that the visual camera **10** can shoot the side surfaces of the plurality of tabs under a certain allowance requirement.

(65) In some embodiments, a maximum height difference H2 between the upper edge of the prism **21** and the lower edge of the prism **21** is greater than a view field B of the visual camera **10**, and the view field B1 of the visual camera **10** is greater than a total thickness h1 of the tab portion **210**, satisfying $H2 > B1 > h1$.

(66) The maximum height difference H2 between the upper edge of the prism **21** and the lower edge of the prism **21** refers to the maximum height difference between the upper edge and the lower edge of the prism **21** in the vertical direction. It can be understood that the height of an uppermost light reflection point on the prism **21** may be located on the upper edge of the prism **21** or lower than the upper edge of the prism **21**; the view field B1 of the visual camera **10** is an inherent parameter of the visual camera **10** and is related to type selection of the visual camera **10**; the total thickness of the tab portion **210** refers to a distance from a highest point of a top tab bulge to a lowest point of a bottom tab recess; and the thickness direction of the tab portion **210** is also the thickness direction of the electrode core **200**, that is, the above first direction.

(67) With the above condition satisfied, the prism **21** can be ensured to reflect the overall side surface of the tab portion **210** and the visual camera **10** can be ensured to shoot the overall side surface of the tab portion **210**.

(68) Referring to FIG. 3 and FIG. 5. in some embodiments, the tab detection mechanism **100** further includes a second drive member **23**, where the second drive member **23** is connected to the prism **21** and configured to drive the prism **21** to move in a direction close to or away from the electrode core **200**.

(69) Specifically, the electrode core **200** is disposed on the electrode core positioning table **310**, and the electrode core **200** is conveyed by the conveying mechanism **300**. The second drive member **23** can drive the prism **21** to move in a direction away from the electrode core **200**, such that the prism **21** is kept away from the electrode core **200** that is being conveyed. When the electrode core **200** is conveyed to the tab detection mechanism **100**, the second drive member **23** drives the prism **21** to move in a direction close to the electrode core **200**, such that the prism **21** is toward the side surface of the tab portion **210**, and then the visual camera **10** is used to acquire an image of the side surface of the tab portion **210**.

(70) With the second drive member **23** provided, the prism **21** is enabled to move in a direction

close to or away from the electrode core **200**, such that the prism **21** is kept away from the electrode core **200** that is being conveyed.

(71) In some embodiments, the second drive member **23** is a cylinder, where the second drive member **23** is configured to drive the prism **21** to move in the vertical direction.

(72) The prism **21** moves in only the vertical direction and can be placed at a side portion of the tab portion **210** after moving to a proper position, thereby ensuring that the prism **21** can reflect light on the side surface of the tab portion **210**.

(73) It can be understood that in other embodiments, the second drive member **23** may also be another servo drive member. Alternatively, the second drive member **23** may drive the prism **21** to move in the horizontal direction, such that the prism **21** is kept away from the electrode core **200** that is being conveyed.

(74) Optionally, the tab detection mechanism **100** includes a prism assembly **20**, where the prism assembly **20** includes a prism **21**, a second bracket **22** and a second drive member **23**. The prism **21** is disposed on the second bracket **22**, and the second bracket **22** is connected to a drive end of the second drive member **23**, so that the second drive member **23** can drive the second bracket **22** and the prism **21** to move in the vertical direction.

(75) In one embodiment, the tab detection mechanism **100** includes a visual camera **10**, a prism assembly **20** and a first drive member **30**. where the visual camera **10** is connected to both a horizontal servo module (not shown in the figure) and the first drive member **30**. A detection process of the tab detection mechanism **100** is as follows: an initial position of the prism **21** is below the conveying mechanism **300**; when the electrode core **200** reaches the tab detection mechanism **100**, the second drive member **23** drives the prism **21** to move up until the tab portion **210** is exactly within the view field of the camera, the visual camera **10** is first driven by the horizontal servo module to position the tab, the first drive member **30** drives the visual camera **10** to move downward, and the visual camera **10** implements multiple times of imaging according to a frequency of 1 mm per time. After imaging is completed, the visual camera **10** and the prism **21** are sequentially returned to the initial position, and the electrode core **200** moves to the next procedure. It can be understood that the horizontal servo module may also be omitted, and in this case, the visual camera **10** is driven by only the first drive member **30** to move in the vertical direction.

(76) Referring to FIG. 3 and FIG. 9, an embodiment of a second aspect of this application discloses a tab detection device **1000** configured to detect a first tab portion **210a** and a second tab portion **210b** on an electrode core **200**. The tab detection device **1000** includes a first tab detection mechanism **100a**, a second tab detection mechanism **100b**, a third tab detection mechanism **100c** and a fourth tab detection mechanism **100d**. At least one of the first tab detection mechanism **100a**, the second tab detection mechanism **100b**, the third tab detection mechanism **100c** and the fourth tab detection mechanism **100d** is the tab detection mechanism **100** according to the first aspect.

(77) The first tab detection mechanism **100a** is configured to detect an inner side surface of the first tab portion **210a**; the second tab detection mechanism **100b** is configured to detect an outer side surface of the second tab portion **210b**; the third tab detection mechanism **100c** is configured to detect an outer side surface of the first tab portion **210a**; and the fourth tab detection mechanism **100d** is configured to detect an inner side surface of the second tab portion **210b**.

(78) The first tab detection mechanism **100a**, the second tab detection mechanism **100b**, the third tab detection mechanism **100c** and the fourth tab detection mechanism **100d** can be arranged in sequence so as to detect the tab portions on the electrode core **200**. The first tab detection mechanism **100a**, the second tab detection mechanism **100b**, the third tab detection mechanism **100c** and the fourth tab detection mechanism **100d** can be arranged in a straight line or in a ring shape, which is not limited in this application. It can be understood that the first tab detection mechanism **100a**, the second tab detection mechanism **100b**, the third tab detection mechanism **100c** and the fourth tab detection mechanism **100d** may not be arranged in sequence, provided that the first tab portion **210a** and the second tab portion **210b** can be detected in sequence.

(79) The tab detection device **1000** includes the first tab detection mechanism **100a**, the second tab detection mechanism **100b**, the third tab detection mechanism **100c** and the fourth tab detection mechanism **100d** that are capable of detecting the inner side surface of the first tab portion **210a**, the outer side surface of the first tab portion **210a**, the inner side surface of the second tab portion **210b** and the outer side surface of the second tab portion **210b**, respectively. The tab detection device **1000** can implement automatic and comprehensive detection of tab portions without manual operation, thereby improving the detection efficiency and accuracy and improving the quality of batteries.

(80) Referring to FIG. 8 to FIG. 10, in some embodiments, a prism **21** in the second tab detection mechanism **100b** and a prism **21** in the third tab detection mechanism **100c** are located on a same second bracket **22** and are arranged opposite to each other. A prism **21** in the first tab detection mechanism **100a** and a prism **21** in the fourth tab detection mechanism **100d** are arranged in a mirror-image form.

(81) The second tab detection mechanism **100b** and the third tab detection mechanism **100c** are respectively used for detecting the outer side surfaces of the second tab portion **210b** and the first tab portion **210a**, the two prisms **21** are separately arranged on two opposite sides of the electrode core **200** and toward the outer side surfaces of the second tab **210b** and the first tab portion **210a**. Therefore, the two prisms **21** can be arranged opposite to each other on the same second bracket **22**.

(82) The prism **21** in the first tab detection mechanism **100a** and the prism **21** in the fourth tab detection mechanism **100d** are arranged in a mirror-image form, so as to detect the outer side surfaces of the first tab portion **210a** and the second tab portion **210b**, respectively.

(83) Further, the second drive member **23** can drive the prism **21** to move in the vertical direction. Therefore, the prism **21** in the second tab detection mechanism **100b** and the prism **21** in the third tab detection mechanism **100c** can share the second bracket **22** and the second drive member **23**, that is, the second drive member **23** is connected to the second bracket **22** to drive the two prisms **21** to move synchronously, simplifying the drive structure and lowering the device cost. In addition, the second tab detection mechanism **100b** and the third tab detection mechanism **100c** can implement synchronous detection, further improving the detection efficiency.

(84) Referring to FIG. 9, FIG. 10 and FIG. 12, in some embodiments, the tab detection device **1000** further includes a light shield plate **40** and a third drive member **50**, where the third drive member **50** is configured to drive the light shield plate **40** to move so that the light shield plate **40** is located between the prism **21** in the second tab detection mechanism **100b** and the prism **21** in the third tab detection mechanism **100c**, and the light shield plate **40** is a non-transparent plate. The third drive member **50** may be, but is not limited to, a cylinder, and may also be a ball screw or another drive member.

(85) The light shield plate **40** can shield light, so as to prevent the prism **21** from reflecting light to a visual camera **10** of an adjacent detection mechanism. The light shield plate **40** may be disposed in the vertical direction, and the light shield plate **40** can extend from a position above the electrode core **200** to a lower end of the telecentric lens **13**.

(86) The second tab detection mechanism **100b** and the third tab detection mechanism **100c** can implement synchronous detection. Therefore, in order to avoid mutual interference of the two detection mechanisms, a movable light shield plate **40** is provided in the embodiment of this application. During detection, the third drive member **50** enables the light shield plate **40** to be located between the second tab detection mechanism **100b** and the third tab detection mechanism **100c**, so that mutual interference is avoided, and the imaging quality will not be affected. After detection, the third drive member **50** enables the light shield plate **40** to be moved out from the position between the second tab detection mechanism **100b** and the third tab detection mechanism **100c**, so as not to hinder the conveying of the electrode core **200**.

(87) With the third drive member **50** and the light shield plate **40** provided, mutual interference

between the second tab detection mechanism **100b** and the third tab detection mechanism **100c** can be avoided, and thus the imaging quality is guaranteed.

(88) In some embodiments, the light shield plate **40** is a non-transparent plate subjected to matte black processing. The light shield plate **40** is subjected to matte black processing, therefore the light shield plate **40** can not only shield light but also avoid light reflection, thereby preventing reflected light from affecting an imaging effect of the visual camera **10**.

(89) In some embodiments, the tab detection device **1000** further includes a conveying mechanism **300**, where the conveying mechanism **300** is configured to convey the electrode core **200**, so that the electrode core **200** sequentially passes through the first tab detection mechanism **100a**, the second tab detection mechanism **100b**, the third tab detection mechanism **100c** and the fourth tab detection mechanism load.

(90) The conveying mechanism **300** may be a conveying belt, a conveying chain, a turnplate, and the like. The type of the conveying mechanism **300** is not limited in this application. The electrode core **200** is disposed on the conveying mechanism **300** through an electrode core positioning table **310**, so that the conveying mechanism **300** can drive the electrode core positioning table **310** and the electrode core **200** to move, thus the electrode core **200** sequentially passes through the four tab detection mechanisms. In this way, the inner and outer side surfaces of the first tab portion **210a** and the second tab portion **210b** are detected, and wrinkling, folding, cracking and the like of the tabs, as well as staggering situation and evenness of a plurality of tab layers are detected.

(91) The tab detection device **1000** further includes light sources (not shown in the figure). Optionally, the first tab detection mechanism **100a**, the second tab detection mechanism **100b**, the third tab detection mechanism **100c** and the fourth tab detection mechanism **100d** each fit with one light source for use.

(92) In conclusion, it should be noted that the foregoing embodiments are for description of the technical solutions of this application only rather than for limiting this application. Although this application has been described in detail with reference to the foregoing embodiments, persons of ordinary skill in the art should appreciate that they can still make modifications to the technical solutions described in the foregoing embodiments or make equivalent replacements to some or all technical features thereof; these modifications or replacements do not separate the essence of the corresponding technical solutions from the scope of the technical solutions of the embodiments of this application, and they shall fall within the scope of claims and specification of this application. In particular, as long as there is no structural conflict, the various technical features mentioned in the embodiments can be combined in any manner. This application is not limited to the specific embodiments disclosed in this specification but includes all technical solutions falling within the scope of the claims.

Claims

1. A tab detection mechanism, configured to detect a tab portion on an electrode core, comprising: a visual camera; a prism, disposed below the visual camera and configured to reflect light on a side surface of the tab portion to the visual camera; and a first driver, connected to the visual camera to drive the visual camera to move in a vertical direction, wherein the visual camera is capable of shooting a plurality of images while moving, wherein the tab detection mechanism further comprises a second driver, and the second driver is connected to the prism to drive the prism to move in a direction close to or away from the electrode core.
2. The tab detection mechanism according to claim 1, characterized in that the visual camera comprises a first bracket connected to the first driver, as well as a camera and a telecentric lens that are disposed on the first bracket, and the telecentric lens and the camera are sequentially disposed above the prism in the vertical direction.
3. The tab detection mechanism according to claim 2, characterized in that an included angle

between the prism and the vertical direction is 45° .

4. The tab detection mechanism according to claim 3, characterized in that a mounting height of the telecentric lens from the prism is greater than or equal to a difference between a working distance of the telecentric lens and a horizontal distance from an upper edge of the prism to the tab portion.
 5. The tab detection mechanism according to claim 3, characterized in that the tab portion comprises a plurality of tabs stacked in a first direction, a moving distance of the visual camera in the vertical direction is greater than a maximum offset distance of the plurality of tabs in a second direction, and the second direction is perpendicular to the first direction.
 6. The tab detection mechanism according to claim 3, characterized in that a maximum height difference between the upper edge of the prism and a lower edge of the prism is greater than a view field of the visual camera, and the view field of the visual camera is greater than a total thickness of the tab portion.
 7. The tab detection mechanism according to claim 1, characterized in that the second driver is a cylinder, and the second driver is configured to drive the prism to move in the vertical direction.
 8. A tab detection device, configured to detect a first tab portion and a second tab portion on an electrode core, comprising: a first tab detector to detect an inner side surface of the first tab portion; a second tab detector to detect an outer side surface of the second tab portion; a third tab detector to detect an outer side surface of the first tab portion; and a fourth tab detector to detect an inner side surface of the second tab portion, wherein at least one of the first tab detector, the second tab detector, the third tab detector and the fourth tab detector is the tab detection mechanism according to claim 1.
 9. The tab detection device according to claim 8, characterized in that a prism in the second tab detector and a prism in the third tab detector are located on a same second bracket and are arranged opposite to each other.
 10. The tab detection device according to claim 8, characterized in that a prism in the first tab detector and a prism in the fourth tab detector are arranged in a mirror-image form.
 11. The tab detection device according to claim 8, characterized in that the tab detection device further comprises a light shield plate and a third driver, the third driver is configured to drive the light shield plate to move, such that the light shield plate is located between the prism of the second tab detector and the prism of the third tab detector, and the light shield plate is a non-transparent plate.
 12. The tab detection device according to claim 11, characterized in that the light shield plate is a non-transparent plate subjected to matte black processing.
 13. The tab detection device according to claim 8, characterized in that the tab detection device further comprises a conveyer to convey the electrode core, so that the electrode core sequentially passes through the first tab detector, the second tab detector, the third tab detector and the fourth tab detector.
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